



Variational auto-encoders based on the shift correction for imputation of specific missing in multivariate time series

Junying Li^a, Weijie Ren^b, Min Han^{c,d,*}

^a The Faculty of Electronic Information and Electrical Engineering, Dalian University of Technology, Dalian 116024, China

^b College of Intelligent Systems Science and Engineering, Harbin Engineering University, Harbin 150001, China

^c Key Laboratory of Intelligent Control and Optimization for Industrial Equipment of Ministry of Education, Dalian University of Technology, Dalian 116024, China

^d Chinad Professional Technology Innovation Center of Distributed Control for Industrial Equipment of Liaoning Province, Dalian University of Technology, Dalian 116024, China

ARTICLE INFO

Keywords:

Missing data
Imputation
Variational auto-encoders
Shift correction
Multivariate time series

ABSTRACT

Data missing is a ubiquitous phenomenon in multivariate time series analysis because of failure measurements, improper installation or other unfavorable factors. Most imputation methods, such as statistical methods and machine learning methods, are mainly for dealing with random or continuous missing, but do not take into consideration the non-random specific missing situation. This paper proposes an imputation model based on the variational auto-encoders (VAE) and shift correction for specific missing values, which is also extended to the β -VAE model. The shift correction is used to correct the original probability distribution deviation caused by specific values concentrated missing. Then taking the meteorological and air quality data sets in Beijing as the example, this paper mainly explores two common specific missing situations, including large values missing and small values missing. The experimental results show that the proposed model can effectively impute specific missing values, which has high imputation accuracy and robustness.

1. Introduction

1.1. Research background

Time series data plays a vital role in the real world, which contains rich information related to temporal dynamics. There is no doubt that if we can dig out effective information to build a forecasting model with high precision, and it will be of practical significance to provide theoretical support for the control and decision in different fields [1–4]. In addition, the quality and accuracy of the time series data obtained from measurements is crucial for the reliability of forecasting model. For example, when some data is missing due to the failure of air quality monitoring sensor equipment, if the data containing missing values is directly used for future prediction, the forecasting results will inevitably cause large errors. In practical application, time series data also usually contains missing values due to failure measurements, transmission error of the measurement system, improper installation and so on. With the increase of missing rate, the forecasting error of the constructed model will also increase. And there even is producing wrong forecasting

results, which will have a great impact on the subsequent analysis and decision [5–6]. Therefore, it is of great crucial to analyze the time series containing missing values and try to impute them through theoretical methods to minimize the impact of measurements or system failure.

1.2. Related works

Data missing has planned their effective use for a long time, and therefore it is also a problem worthy of study in the field of time series and extensive studies have been conducted to this problem. Ref. [7] presented that the main imputation methods for data missing can be divided into the simplest deletion method, simple linear imputation method based on statistics and nonlinear imputation method based on machine learning. Among them, the deletion method directly discards missing values. However, this method may cause a large deviation in the analysis results due to the loss of important information [8–9]. The simple linear imputation method based on statistics is to replace missing values by the average [10], median [11–12] or the most common value [13]. Furthermore, according to the statistical characteristics of the

* Corresponding author at: Key Laboratory of Intelligent Control and Optimization for Industrial Equipment of Ministry of Education, Dalian University of Technology, Dalian 116024, China.

E-mail address: minhan@dlut.edu.cn (M. Han).

<https://doi.org/10.1016/j.measurement.2021.110055>

Received 2 May 2021; Received in revised form 15 August 2021; Accepted 17 August 2021

Available online 11 September 2021

0263-2241/© 2021 Published by Elsevier Ltd.

obtained data, complex linear regression can also be used to impute, such as expectation maximization (EM) [14], k-nearest neighbor (KNN) [15], matrix factorization [16], multivariate imputation chain equation (MICE) [17] and so on. At the same time, in order to facilitate the processing of missing data, many researchers have developed the integrated toolkits for imputing missing data. For example, the Autoimpute Python package integrates common imputation methods such as mean, MICE, linear regression and so on. However, these statistical methods do not take into account the correlation and dependence between variables and data of time series. And most statistical imputation methods are often based on the linear hypothesis, which is not conducive to the popularization and application of the actual nonlinear time series with data missing.

Therefore, in recent years, with the development of machine learning and deep neural network, there are many integrated toolkits and imputation methods for data missing. For toolkits, the DataWig toolkit is based on deep learning modules and neural architecture search [18], and, and the two-step verifiable imputation process is aimed at different missing situations of univariate time series [19]. And for imputation methods, with the powerful nonlinear mapping ability and fast learning speed, the generalized regression neural network (GRNN) has been applied to the missing data management of the actual intelligent analysis system [20–23]. Auto-encoders (AE), as a self-supervised learning model, forming a framework for missing value imputation because of its function of generating samples similar to the training data. Therefore, it is widely used in data augmentation and data imputation, and which has achieved great results in genome data, image data and industrial data [24–27]. However, for the standard AE model, there are no constraints in the training process. It only passively copies the input and creates samples close to the training data, which is lack of learning the potential characteristics and probability distribution of the data. Therefore, when the input is new or incomplete, the imputation effect of the AE model is poor [28–29].

The denoising auto-encoders (DAE) model is a kind of auto-encoder that uses noise to destroy input and create robust latent features, which has been widely used in imputation due to its strong robustness [30–32]. Besides, as an improved probability learning model based on the AE model, the variational auto-encoders (VAE) model can produce samples similar to the input by imposing regularization constraints to the latent space to learn the probability distribution of the input. By constraining the probability distribution of the latent space to be close to the standard Gaussian distribution, a smooth distribution can be obtained, which enables the VAE model to give meaningful reconstruction for any point sampled in the latent space [33]. And it has been confirmed to be able to express the data imputation as a latent variable model [34]. Compared with the standard AE and DAE model, the VAE model has the merits of stable training, strong interpretable and robustness to outliers [35–38], so it has gradually become one of the main imputation models to deal with nonlinear time series with data missing in recent years [34,38–40].

1.3. Problem and contribution

However, the current imputation methods based on the VAE model often assume that the distribution after data missing is consistent with that of the original data, and impute missing data by learning the “good” values of samples, which does not consider the impact of data missing on the original probability distribution. For non-stationary time series, such as meteorological data with periodicity and air quality data related with seasons, when data missing is concentrated in a specific period, the original probability distribution will change greatly, and the imputation performance will decrease accordingly. Furthermore, the regularization degree of latent space of the VAE model also has a greater impact on imputation results. Imputation results of the standard VAE model is often fixed around a value that minimizes the loss function, which may not match the ground truth. To solve above problems, this paper proposes a shift correction VAE (SC-VAE) model, and generalizes it to the

β -VAE model (shift correction β -VAE, SC- β -VAE). Based on the β -VAE model, it corrects the probability distribution of off-set by shift correction to improve the imputation performance in the case of specific data missing.

And the contribution of this paper can be summarized as follows:

- (1) The basic assumption of the standard VAE model is modified by the shift correction. For the standard VAE model, the basic assumption that the training data and the test data obey the same distribution will produce a large deviation when the data is centralized missing. In this paper, the shift correction is used to correct the deviated distribution to make up for the influence of this deviation.
- (2) A SC- β -VAE imputation model is proposed. The shift correction is extended to the β -VAE model to propose the SC- β -VAE model. This model regulates a trade-off between the ability of generation and disentanglement through the hyperparameter β . And the shift correction improves the imputation accuracy of the model, which is specially used to solve the problem of specific missing in nonlinear multivariate time series.
- (3) The influence of hyperparameters and noises on imputation performance is discussed. By setting different degree of latent spaces regularization, shift correction hyperparameter and different ways of noise introduction in experiments, this paper further observes and analyzes their impact on the imputation accuracy, so as to determine the best training mode.

The other parts of this paper are arranged as follows. Section 2 briefly introduces the basic theoretical background of the VAE model, and describes problems of existing imputation models in detail. Section 3 introduces the proposed method in detail. After this, we discuss the experimental results in Section 4. Finally, we draw the conclusion and future studies in Section 5.

2. Preliminary and problem formulation

This section mainly introduces the principle of generating samples from the VAE model and describes the practical problems to be solved in this paper in detail.

2.1. Variational auto-encoder

The VAE model is a typical deep generative network first proposed by Kingma et al. [41] in 2014, which is a generative network structure based on variational Bayesian inference. It mainly includes two parts, namely inferential network and generative network, which is both customized deep neural network s with nonlinear mapping functions. Just as shown in Fig. 1, the structure of the standard VAE model is

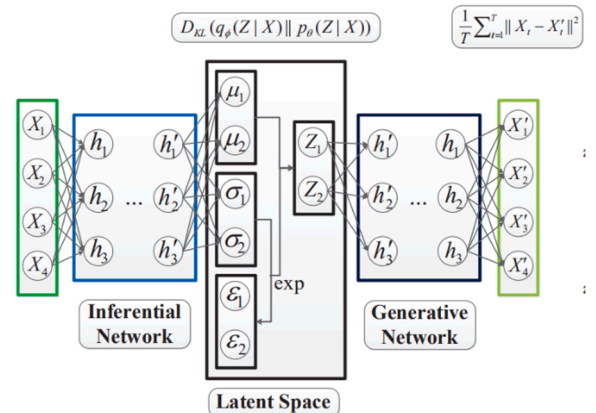


Fig. 1. The network architecture of the standard VAE model.

illustrated. The inferential network is equivalent to an encoder, which is used to perform variational inference on the input data to generate a variational probability distribution of latent factors. And the generative network is equivalent to a decoder, which is used to restore the probability distribution of the input by latent factors.

But different from the specific encoding in the AE model, the main idea of the VAE model is to map the complex input into the latent space, and reconstruct the output from the latent space to realize the learning of the original probability distribution. Therefore, both the inferential network and the generative network are constrained by the probability density distribution of variational parameters. The VAE model encodes the input in the inferential network and enters the latent space. And it expresses the input latent factors as the probability distribution constrained by the mean and variance. Then it randomly samples from the latent probability distribution and enters the generative network to decode the latent space, which can output the reconstructions.

Data sets of input continuous observation variables with unknown distribution can be expressed as $X = xtdt = 1T \in RT \times D$. And for each $xt dt \in 1, 2, \dots, T, d \in 1, 2, \dots, D$, it represents the observed value of the d -th dimension at time t . Let reconstruction sets approximate to the original probability distribution generated by the model be expressed as $X' = xt' dt = 1T$. And let $Z = z(n) n = 1N$ be a vector set of continuous random latent variables, which defines a low dimensional representation that varies significantly with variable X . The latent vector Z can fully represent the whole input composed of variables X , which is a comprehensive distribution similar to Gaussian mixture distribution. According to the probability of different samples, sampling from the distribution of latent vector one by one can reproduce the original data. The prior of latent variable is the multivariate Gaussian distribution with isotropic center.

From the perspective of probability distribution, the standard VAE model mainly includes two likelihood distributions. The first one is the unknown distribution of the input data. It can be generated through latent continuous random vector Z and conditional distribution $p\theta(x'|z)$ and $p\theta(z)$, where θ is the parameter of neural network. And θ can be fitted by maximizing the marginal likelihood of the input [42], as shown in Eq. (1):

$$\theta = \operatorname{argmax}_{\theta} \sum_t = 1T \log \int p\theta(xt, zt) dz \quad (1)$$

The second one is the posterior distribution of input data $p\theta(z|x)$. Here a recognition model $q\varphi(z|x)$ is introduced to replace the above distribution that cannot be determined. According to the above likelihood distributions, the VAE model can learn the probability distribution of input with the constraints of auxiliary parameters that are generated by the deep neural network. And it continuously optimizes the evidence lower bound (ELBO) through the objective function, so as to generate the reconstructions approximate to the original probability distribution [42], as shown in Eq. (2):

$$ELBO = -DKL(q\varphi(z|x)||p\theta(z)) + E_{q\varphi(z|x)}[\log p\theta(x|z)] \quad (2)$$

Since the latent vector Z cannot be obtained by direct observation or maximum likelihood estimation, the actual posterior distribution cannot be determined. Therefore, through the probability distribution $q\varphi(z|x)$, the standard VAE model approximates the real posterior distribution $p\theta(z|x)$, then the probability distribution of the generative network is expressed as $p\theta(x'|z)$. In order to make the probability distribution of the inferential model close enough to the real posterior distribution, the first term in Eq. (2) uses KL divergence to describe the similarity between the two distributions. It is called regularization term and used to control the ability of disentanglement, that is, the ability of the model to separate the mixed multi-dimensional latent space. The second term describes the similarity between the generated samples and the original data, which is called reconstruction term. It is used to control the ability of generation, that is, the ability of the model to sample from the learned probability distribution and reconstruct imputation samples.

2.2. Problem description

The basic assumption of the standard VAE model is that the training data follows the same distribution as the test data. When the VAE model learns the original probability distribution from the training data, the output generated to replace missing values will follow the similar distribution by default. However, in practical application, for time series data with obvious periodicity, when the measurement equipment fails, data missing may not be random, but be concentrated in a certain period. Then, compared with the case of random missing, the probability distribution of the time series containing missing values will obviously change. And there is a greater deviation compared with the original probability distribution, so the imputation performance of the VAE model will decrease. For example, meteorological data or air quality data are affected by seasons, weather changes and so on. July and August in the northern hemisphere belongs to summer, the temperature is significantly higher than the annual average, and the average daily concentration of PM2.5 or other pollutants is lower than the annual average. When the missing of these two data sets is concentrated in summer, the missing of temperature variable is called large values missing situation, and the missing of PM2.5 concentration variable is called small values missing situation. The actual probability distribution of these two kinds of specific missing situation will deviate greatly from the original probability distribution.

Additionally, when the standard VAE model is applied to imputation, its latent space is constrained by the regularization of Gaussian distribution. In order to minimize the loss function as much as possible, the generated output generally floats up and down in a fixed value and often do not fully match missing values, which is not conducive to widespread promotion and application. Ref. [43] pointed out that the hyperparameter β can be introduced to find the explainable factorization latent vectors. And they introduced β into the loss function of the standard VAE model to balance the reconstruction loss term and the regularization loss term, which is called the β -VAE model. By changing the regularization degree of the latent space, the ability of generation can be well balanced, which effectively avoids generated samples concentrating near the fixed value. This idea has been confirmed to be superior to the standard VAE model in some tasks, which attracts more and more research interests [44].

The variation of hyperparameters controls the ability of generation and the latent space's ability of disentanglement. The research results show that when the hyperparameter β is less than 1, and then the

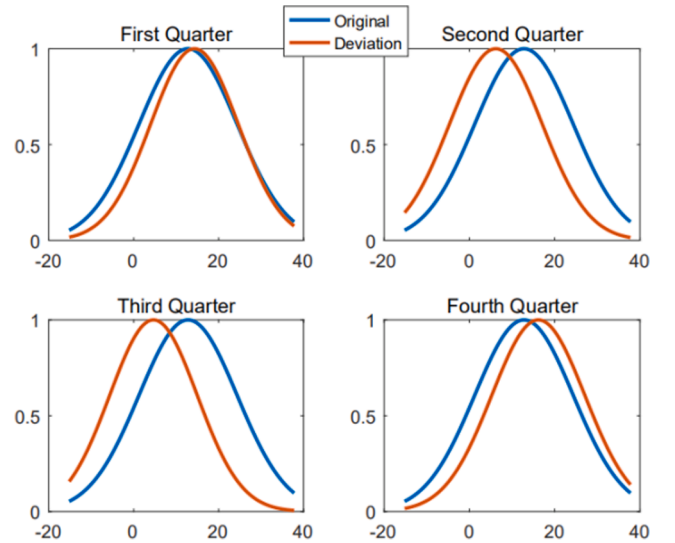


Fig. 2. The original probability distribution and the probability distribution containing missing values in different specific missing situations (taking the temperature as an example).

reconstruction effect is better than the standard VAE model. In the field of image generation, the influence of hyperparameters on experimental results has attracted more and more researchers [44]. However, the effect of the changeable β on the imputation performance has not been extensively explored and widely studied. In addition, the AE model lacks noise training, the DAE model uses noise to destroy the input and latent space of the VAE model introduces noise constraints. Different introduction of noise will have different impact on the imputation results. Unreasonable regularization constraints of latent space or inappropriate introduction of noise both may be factors that cause performance degradation.

3. Methodology

In order to solve the imputation problem of specific data missing in practical application, this paper proposes an improved VAE model based on shift correction and extends it to the β -VAE model, namely SC- β -VAE. This section mainly introduces the applicable principle of the SC- β -VAE model in specific data missing.

As shown in Section 2.2, large values missing situation and small values missing situation are common missing types in time series with obvious periodicity, such as meteorological or air quality data. For example, in the meteorological data set, when most of missing values of temperature are concentrated in the weather with high temperature in summer, we can basically think that the original probability distribution will shift to the direction of smaller values. As shown in Fig. 2, taking the temperature as an example, the Gauss distribution change of the latent space's map in different seasons is drawn. Obviously, in the second and third quarter, the probability distribution including missing values shifts to the smaller value. And in the first and fourth quarter, the probability distribution shifts to the larger value. On the contrast, in the air quality data set, when most missing values of PM2.5 or other pollutants' average daily concentration are concentrated in summer, due to the impact of seasonal change and social activities, missing values are mostly small. Then, we basically think that the original probability distribution will shift to the direction of larger values. In Section 2.1, the basic assumption of the standard VAE model is that the training data and test data follow the same probability distribution. In other words, the standard VAE model learns the probability distribution from the training data, and constrains the latent vector Z to follow a Gaussian distribution

by introducing noise. The output is generated by decoding the latent vector and follows the probability distribution that is exactly consistent with the training data by default. However, in specific missing situation, data missing shifts the original probability distribution, which no longer meets the basic assumptions of the standard VAE model, and the imputation performance decreases accordingly. Therefore, this paper proposes the shift correction to correct the basic assumptions in order to better estimate imputation samples to replace missing values.

For the standard VAE model, the latent vector Z follows the Gaussian distribution $Z \sim N(\mu, \sigma)$. In the case of large values missing situation, as shown in Fig. 2, the probability distribution learned by the model has a smaller mean than the original probability distribution, so the reconstructed samples are also smaller than the real data. On the contrary, in the case of small values missing situation, the reconstructed samples have larger values. In order to make the imputation samples closer to the real data, this paper modifies the assumption of the input probability distribution, that is, to constrain the latent vector Z to follow the Gaussian distribution with shift correction $Z \sim N(\mu + \lambda\sigma, \sigma)$. Among them, μ and σ are still the mean and variance of Z respectively, and λ is called shift hyperparameter. By shifting the mean from μ to $\mu + \lambda\sigma$, it makes the model also following the basic assumption as much as possible when learning input probability distribution containing missing values.

At the same time, in order to get better imputation effect, we extend the improved model to the β -VAE model. Through the control of regularization constraints of latent space, we can further determine the effect of the regularization on the performance improvement for specific data missing imputation. Then the loss function of the standard VAE model is modified as shown in Eq. (3):

$$Loss = -EQ\varphi(z|x)[\log P\theta(x|z)] + \beta DKL(Q\varphi(z|x)||P\theta(z)) \quad (3)$$

To sum up, the schematic diagram of the SC- β -VAE model is shown in Fig. 3. Where, $Z \sim N(\mu + \lambda\sigma, \sigma)$ and μ, σ are both generated from neural networks controlled by θ and φ , which represents the mean and variance of the original probability distribution of the latent vector respectively. The hyperparameter β is used to control and balance the ability of generation and disentanglement and the hyperparameter λ is used to control the shift correction of probability distribution. For the hyperparameter β , when $\beta = 1$, the VAE model degenerates to the standard VAE model. When $\beta > 1$, it means that the regularization degree of the latent space increases, so the probability distribution of latent space is

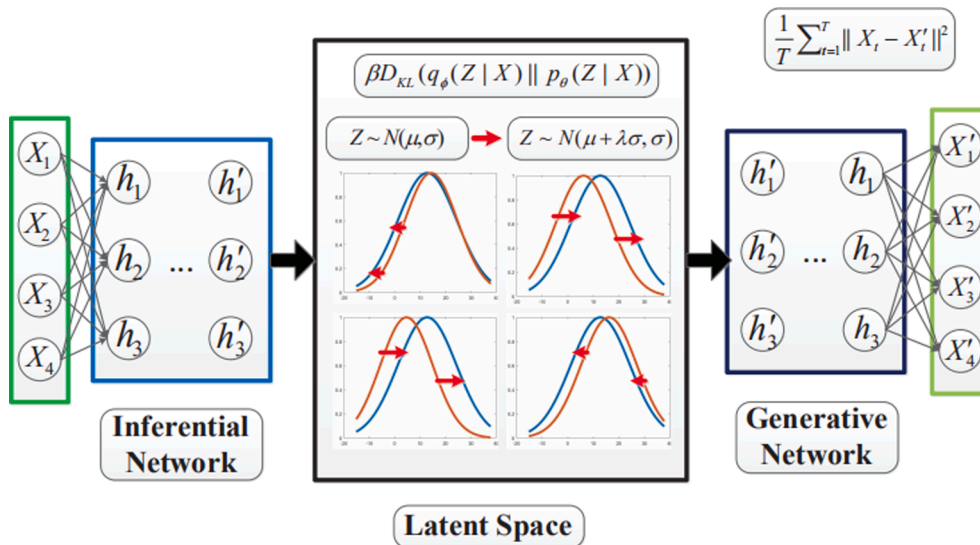


Fig. 3. Schematic diagram of the SC- β -VAE model.

smoother and its ability of disentanglement is stronger. Then the model has higher coding efficiency but its ability of generation decreases accordingly. In contrast, when $\beta = 0$, the regularization term is removed and the loss function only contains the reconstruction term. The β -VAE model will degenerate to the ordinary AE model without any regularization. When $0 < \beta < 1$, the β -VAE model may find the balance between the reconstruction term and the regularization term, so it has a better ability of generation and disentanglement at the same time. For the hyperparameter λ , in practical application, because the influence of large or small values missing on the original probability distribution is unknown, it is difficult to precisely control the determination of the hyperparameter λ . This paper generally simulates different types and missing rates of the masking to calculate and select the corresponding range of the hyperparameter λ .

As shown in Section 2.1, data sets with missing values can be expressed as $X = xtdt = 1T \in RT \times D$. The random latent vector and output are expressed as Z and $X' = xt'dt = 1T$ respectively. Each xt in data sets has the possibility of missing. For specific missing situation, the missing probability of xt at the specific period will be much higher than that of other periods. As described in Section 2.1, the latent vector Z is determined by the distribution generated by the inferential network and the generative network. These two networks are constrained by parameters ϕ and θ , and their gradient optimization is expected by minimizing KL divergence, as shown in Eq. (4):

$$\theta, \phi = \operatorname{argmin}_{\theta, \phi} DKL(q\phi(z|x) || p\theta(z|x)) \\ = Eq\phi(z|x)[\log q\phi(z|x) - \log p\theta(z|x)] + \log p\theta(x) \quad (4)$$

For the convenience of expression, let $L(\theta, \phi; x) = -Eq\phi(z|x)[\log q\phi(z|x) - \log p\theta(z|x)] = Eq\phi(z|x)[\log p\theta(z|x) - \log q\phi(z|x)]$ and substitute it into Eq. (4) and combine Eq. (3) to get Eq. (5):

$$\log p\theta(x) = DKL(q\phi(z|x) || p\theta(z|x)) + L(\theta, \phi; x) \quad (5)$$

The left side of Eq. (5) is the term we expect to maximize. The first term on the right side of it is KL divergence between the approximate posterior distribution and the true posterior distribution, which is always nonnegative. Therefore, maximizing the left term can be achieved by maximizing the so-called ELBO. Then the equation is changed to that shown in Eq. (6):

$$L(\theta, \phi; X) = Eq\phi(z|x)[-\log q\phi(z|x) + \log p\theta(z|x)] \\ = -\beta DKL(q\phi(z|x) || p\theta(z|x)) + Eq\phi(z|x)\log p\theta(x'|z) \quad (6)$$

To sum up, the optimization objective of the inferential network and the generative network is to maximize the evidence lower bound, so the overall optimization objective of the VAE model is to maximize the ELBO. In order to facilitate the calculation of latent vector Z , an auxiliary parameter with the known likelihood is introduced to transform the distribution $q\phi(z|x)$ of recognition model, and $z = g\phi(\epsilon, x)$ [45]. Here, ϵ is an auxiliary parameter introduced to ensure that the noise variance automatically sampled from the normal distribution is positive. And $g\phi(\epsilon, x)$ is a function to obtain the latent distribution after distribution transformation of $q\phi(z|x)$. When $q\phi(z|x)$ is the normal distribution, the latent vector Z can be sampled by auxiliary parameters as shown in Eq. (7) and $Z \sim N(\mu + \lambda\sigma, \sigma)$.

$$zd = (\mu d + \lambda \cdot \sigma d) + \sigma d \cdot \epsilon d \quad (7)$$

For the latent vector after sampling from $p\theta(z) \sim N(0, 1)$, the calculation of the evidence lower bound is shown in Eq. (8):

$$L(\theta, \phi; X) = -\beta DKL(q\phi(z|x) || p\theta(z)) + 1D \sum d = 1D \log p\theta(x'd|zd) \\ = 12 \sum d \\ = 1D[\beta(\log(\sigma d)^2 - (\mu d + \lambda \cdot \sigma d)^2 - (\sigma d)^2 + 1) + \log p\theta(x'd|zd)] \quad (8)$$

Algorithm 1. (SC- β -VAE for imputation of specific missing situations)

Input: Complete train data set; Incomplete test data set; Appropriate hyperparameters β and λ

Output: Imputed test data set

- 1: Initializing SC- β -VAE as model;
- 2: Feeding complete train data set X as input into initialized SC- β -VAE;
- 3: Calculating new loss function by Eq. (8) and outputting trained SC- β -VAE;
- 4: **for** number of testing epochs **do**
- 5: **for** number of iterations in one epoch **do**
- 6: Feeding minibatch samples X as input from test data set into trained SC- β -VAE and outputting reconstructions X' ;
- 7: **end**
- 8: **end**
- 9: **Return:** Reconstruction X'

After introducing auxiliary parameters, the latent vector Z can be calculated by stochastic gradient optimization. And Algorithm 1 shows the imputation process of the SC- β -VAE in detail. The conditional distribution follows Bernoulli distribution or Gaussian distribution, so the mean and variance of the conditional distribution are calculated by inferential network and generative network. And then they can be calculated by the probability density function formula. In a word, SC- β -VAE learns the input through the inferential network to approximate the posterior distribution of the actual data $p\theta(z|x')$. Then it generates the mean and variance of the latent probability distribution in the latent space. Finally, the latent vector Z is randomly sampled from the learned distribution $q\phi(z|x')$, and the reconstructions $X' = xt'dt = 1T$ are output by the generative network to replace missing values.

4. Case study

In this section, four real public datasets are used to comprehensively evaluate the imputation performance of the proposed model from different perspectives. Additionally, in practical application, the imputation model is unsupervised learning model, so it is impossible to compare directly with the ground truths. Therefore, the experiments of imputation task and prediction task are designed to fully demonstrate the superior performance of the proposed model. Among them, the experiments of hyperparameter, noise, specific missing (missing not at random (MNAR)) and missing completely at random (MCAR) imputation tasks were carried out on the meteorological and air quality datasets in Beijing. Through artificial masking, different missing conditions are set to generate incomplete datasets. And imputation results of the proposed model are compared with the comparison models, which helps to show the great performance of the proposed model more intuitively. In addition, the experiment of prediction task is carried out on two competition datasets KDD CUP 2018 and Physionet Challenge 2012. Taking the training results of different imputation models as the input of the same prediction model, the prediction results are calculated to verify the performance of the imputation models. The model codes for this work and related data used in these experiments can be available under the open-source MIT license and be accessible on GitHub¹.

4.1. Dataset description

- (1) Meteorological dataset of Beijing. This dataset includes historical meteorological data from June 1, 2017 to December 31, 2020 in Beijing. We choose the average measured values of all 12 main monitoring stations in Beijing as the meteorological records of the region and store them in CSV for experiments. Each monitoring station includes records observed every day of 5 meteorological measurement variables (temperature (T), pressure (P), humidity (H), wind speed (WS) and wind direction (WD)). In this

¹ <https://github.com/lh77hl/SC-beta-VAE/branches>

Table 1

Summary of the important optimal hyperparameters set.

Models	Hyperparameters	Types of datasets	The optimal set
β -VAE and SC- β -VAE	β	Mete(1st and 4th) and AQI(2nd and 3rd) with MNAR	0.5
		Mete(2nd and 3rd) and AQI(1st and 4th) with MNAR	0.01
		Mete, AQI and KDD with MCAR	0.05
SC- β -VAE	λ	Mete(1st and 4th) and AQI(2nd and 3rd) with MNAR	1.0
		Mete(2nd and 3rd) and AQI(1st and 4th) with MNAR	5.0
		Mete, AQI and KDD with MCAR	4.0
KNN	k	Mete	5
		AQI	7
		Phy	15
DAE and SC-DAE	noise factor	Mete and AQI	0.25

Table 2Imputation results (in RMSE) of specific missing situations on different data sets with different β . (The top is the meteorological data, and the bottom shows the air quality data).

Quarters	0.0	0.01	0.05	0.1	0.5	1.0	5.0	10.0
First Quarter	0.2486	0.2481	0.2356	0.2209	0.1458	0.1653	0.1959	0.2153
	0.6919	0.6456	0.6489	0.6544	0.6551	0.6555	0.6578	0.6660
Second Quarter	0.2698	0.1734	0.1949	0.2258	0.2482	0.2487	0.2505	0.2517
	0.6092	0.5713	0.5655	0.5483	0.5378	0.5921	0.5992	0.5988
Third Quarter	0.3559	0.2436	0.2605	0.2858	0.3036	0.3046	0.3194	0.3200
	0.4627	0.4550	0.4424	0.4415	0.4272	0.4556	0.4560	0.4557
Fourth Quarter	0.2460	0.2460	0.2445	0.2373	0.1629	0.1802	0.2066	0.2176
	0.6679	0.6425	0.6436	0.6499	0.6513	0.6526	0.6602	0.6623

paper, the whole data set is divided into train and test set according to the ratio of 7:3. The train set is fed to the model during the training, and the test set is used to verify the performance of the training model. We split the above dataset for every 21 days and perform two kinds of imputation tasks as follows: 1) The imputation task for specific missing situation (missing not at random (MNAR)). Taking the quarter as a unit, for every 21 days length sample in each specific quarter, 50% of the data is randomly discarded to simulate the centralized missing. Samples in other quarters remain unchanged, and then different types of specific missing situation are generated. Then the model imputes missing values and the imputation accuracy is defined as the rootmean square error (RMSE) between the ground truths and imputation values. 2) The imputation task for missing completely at random (MCAR). Different from the specific missing situation, we randomly discard 50% for every 21 days of the whole dataset to fully simulate the MCAR. Then the missing values are imputed, and the imputation accuracy is still defined as RMSE.

(2) Air quality dataset of Beijing. This dataset includes historical air quality data from January 1, 2015 to December 31, 2020 in Beijing. We choose the average measured values of all 12 main monitoring stations in Beijing as the air quality records of the region and store them in CSV for experiments. Each monitoring station includes records observed every day of 7 air quality measurement variables (AQI, PM2.5, PM10, SO₂, NO₂, O₃ and CO). The division of dataset and imputation task are consistent with the meteorological dataset.

(3) Physionet Challenge 2012 dataset. The Physionet dataset is a public electronic medical record dataset from the PhysioNet Challenge 2012 [46], which is a task for predicting whether a patient will die in hospital or not. The dataset consists of 4000 ICU inpatient records, each of which lasts 48 h and contains 41 variables such as age, weight and heart rate. Because there are missing values in the dataset, this paper uses the same classifier SVM to predict the mortality results based on the proposed model and comparison models. It can verify the imputation performance

of different models and define the prediction accuracy as AUC score of binary classification problem.

(4) KDD CUP 2018 Dataset. KDD CUP 2018 dataset is the public air quality dataset from KDD CUP Challenge 2018². This paper selects the historical air quality data of London in 2017, and a total of 12 variables were recorded every day at each monitoring station such as PM2.5, PM10, CO, weather, temperature and so on. The segmentation of train and test set is consistent with that of Beijing dataset. The above dataset is divided every 48 h, and we conduct the prediction task as the following described. We sample every 48 h in each quarter and randomly discard 50% of the dataset, which keeps data in other quarters unchanged and generates different types of specific missing situation. After the imputation task is completed by the proposed model and the comparison models, we predict the average air quality in the next 6 h and calculate the prediction results of air quality to determine the performance of imputation methods. The prediction accuracy is defined as the mean square error (MSE) between the ground truths and imputation values.

4.2. Experiment settings

4.2.1. Baseline methods for comparison

In order to better show the excellent performance of the proposed model, five kinds of comparison models are selected for different experiments.

(1) Models of auto-encoder types. Some baseline models of auto-encoder types are selected for the experiments of noise, specific missing (missing not at random (MNAR)) and missing completely at random (MCAR). Among them, the auto-encoders (AE) model is a typical self-supervised learning model applied to data augmentation. By encoding and decoding, it reconstructs the samples which approximately conforms to the original distribution. The VAE model [41] is an unsupervised learning model and the basic model introduced in Section 2.1. The denoising auto-encoders (DAE) model [47] is different from the VAE model

² https://www.biendata.xyz/competition/kdd_2018/

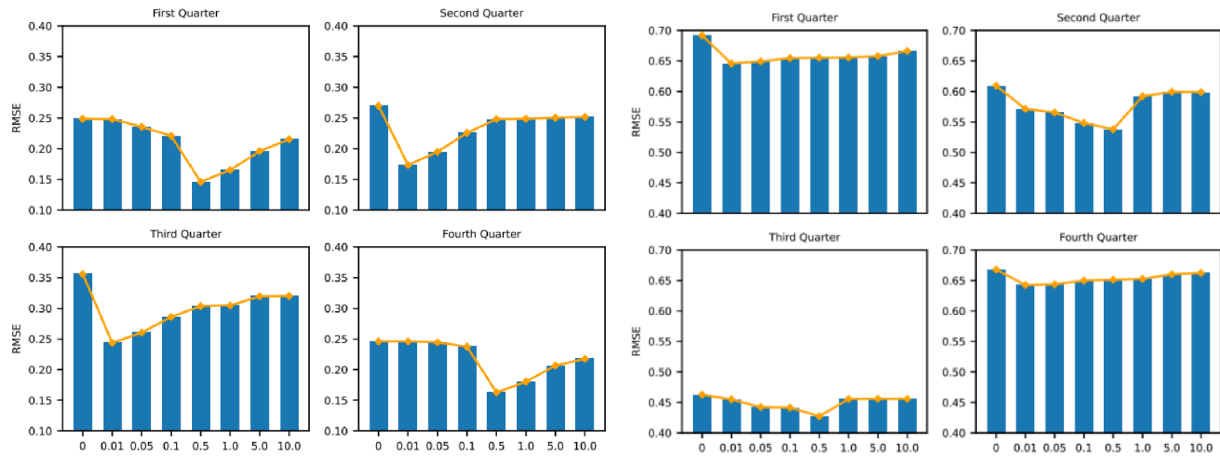


Fig. 4. RMSE performance of specific missing situations on the meteorological data with different β .

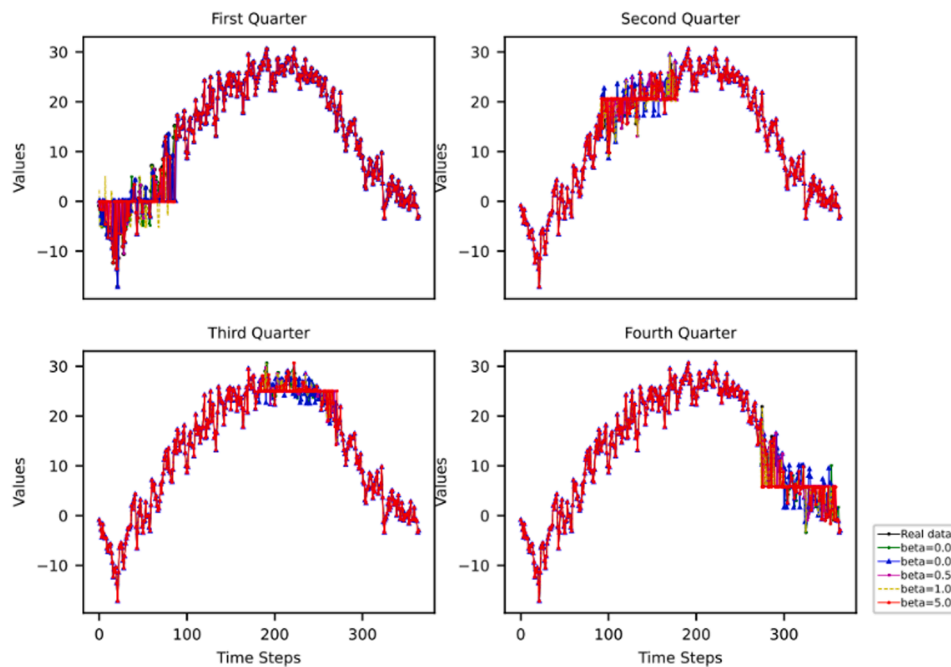


Fig. 5. RMSE performance of specific missing situations on the air quality data with different β .

trained with noise injected in its stochastic hidden layers. It is trained to reconstruct the input data with noise injected at the input level, whose latent space has better average log-likelihood than the VAE model when the input data is missing.

- (2) Statistical methods. Some typical statistical imputation methods are selected for the comparison of experiments under specific missing conditions. Among them, mean imputation is one of the most fundamental statistical imputation methods, which uses the global mean of historical observations to impute missing data. Another method is Softimpute used for large scale matrix data missing [48], which is an algorithm to impute missing data through soft threshold processing of singular value decomposition. The main method of KNN is the mean square error of adjacent samples weighting with high similarity, and weighted results impute missing values [15].
- (3) Machine learning methods. Some typical machine learning imputation methods are selected for the comparison of experiments under specific missing conditions. Among them, random forest uses each feature with missing value as a label to estimate and impute. Iterative models [17] each feature with missing

value as a function of other features in a circular manner to estimate missing values.

- (4) Generative models. The VAE model is one of the typical representative generative models. Here we choose some baseline models of this kind for comparison. Among them, the generative adversarial network (GAN) [49] is also a typical unsupervised learning model, which can generate samples similar to the original data as imputation results through the game of generative network and discriminant network. The Wasserstein-GAN (WGAN) model [50] is an improved model based on the GAN model and the optimal transport theory. It can effectively solve the gradient disappearance problem of the GAN model by smoother Wasserstein distance, which obtains better imputation results. The adversarial variational bayes model (AVB) [51] establishes the relationship between the VAE model and GAN model, which introduces the auxiliary discriminant network and trains the VAE model by inferential model with arbitrary expressions. Then it can provide the maximum posterior distribution of adjacent variables under given observation values, which ensures that the generated samples are closer to the original data.

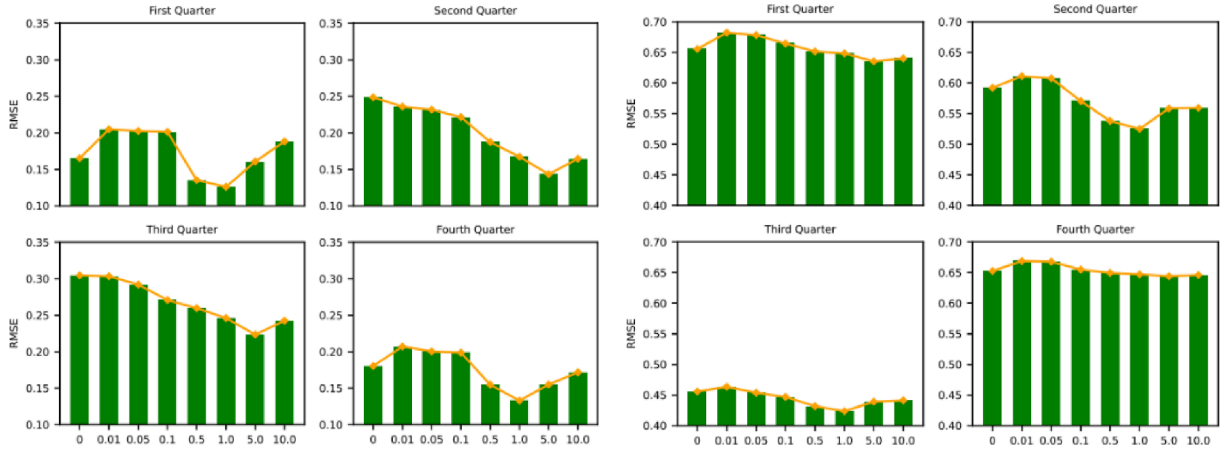


Fig. 6. Imputation results of different β on the meteorological data.

Table 3

Imputation results (in RMSE) of specific missing situations on different data sets with different λ . (The top is the meteorological data, and the bottom shows the air quality data).

Quarters	0.0	0.01	0.05	0.1	0.5	1.0	5.0	10.0
First Quarter	0.1652	0.2049	0.2025	0.2014	0.1993	0.1257	0.1351	0.1607
Second Quarter	0.6555	0.6822	0.6782	0.6644	0.6593	0.6515	0.6483	0.6353
Third Quarter	0.2487	0.2360	0.2317	0.2216	0.2179	0.1874	0.1673	0.1434
Fourth Quarter	0.5921	0.6111	0.6077	0.5705	0.5696	0.5248	0.5378	0.5584
First Quarter	0.3045	0.3033	0.2919	0.2709	0.2665	0.2597	0.2461	0.2235
Second Quarter	0.4555	0.4633	0.4536	0.4469	0.4415	0.4235	0.4316	0.4390
Third Quarter	0.1801	0.2073	0.2004	0.1988	0.1778	0.1329	0.1547	0.1549
Fourth Quarter	0.6525	0.6689	0.6678	0.6548	0.6552	0.6494	0.6471	0.6440

(5) KDD CUP 2018 competition methods. This manuscript selects several excellent algorithms to carry out the experiment of prediction task for comparison from KDD CUP 2018, which further verifies the imputation performance of proposed model through the accuracy in prediction task. They mainly include XGBoost and other modeling methods, which are recorded as KDD1³, KDD2⁴, KDD3⁵.

4.2.2. Implementation details

According to the introduction in Section 3, this paper implements the model in the Tensorflow framework and uses the Adam optimizer. In the implementation of the auto-encoder types, as a compressed space, the dimension size of latent space is often smaller than the input dimension, but there is no exact rule to determine its size. Therefore, this paper uses Bayesian optimization to determine parameters on different data sets. Here, we use a model structure with five hidden layers, and the dimensions of hidden layers adjacent to the latent space have 10-fold size increases. The activation function of hidden layer is set as sigmoid function. Because training results of the model are not sensitive to the change of batch size and learning rate, they are set as 100 and 0.005 in experiments. In addition, in order to present the error and imputation performance of various models conveniently, the experimental results are calculated by standardizing the data to the range of 0 to 1. Additionally, the other important optimal hyperparameters set of the proposed model and comparison models on different experiments are determined as shown in Table 1. (Where, the Meteorological dataset is abbreviated as Mete, air quality dataset is abbreviated as AQI, Physionet dataset is abbreviated as Phy, and different quarters are abbreviated as the corresponding number.)

4.3. Experimental results

4.3.1. Effect of the hyperparameters

A. The hyperparameter β . With the continuously distributed latent vector Z , the standard VAE model can be regarded as a mixed Gaussian model formed by the combination of infinite independent Gaussian distribution with different mean and variance. The probability distribution is entangled, so the model lacks good interpretability. The β -VAE model introduces the hyperparameter β into the loss function and further regulates a trade-off between the ability of generation and the latent space's ability of disentanglement. This subsection studies the effect of $\beta \in \{0.0, 0.01, 0.05, 0.1, 0.5, 1.0, 5.0, 10.0\}$ on the imputation performance of specific missing situation and further observes how the regularization degree of latent space controlled by the hyperparameter β affects the imputation accuracy. We can explore the potential possibility of increasing the imputation accuracy of the VAE model, which is helpful to further understand its mechanism and determine the best hyperparameter.

Table 2 presents results of RMSE in specific missing situation, which shows the effect of different β on the imputation accuracy of meteorological and air quality data respectively. To further intuitively compare experimental results, RMSE is regarded as the ordinate to draw imputation performance figures on the meteorological and air quality data, as shown in Fig. 4 and Fig. 5 respectively. The time stamps in test set of the meteorological data are regarded as an example, and the imputation results of different β are shown in Fig. 6. We observed that when $\beta = 0$, that is, the model degenerates to the standard AE model and the imputation accuracy is the lowest. This is because the AE model lacks the learning of the original potential characteristics and only carries out the ordinary reconstruction. Therefore, when the data is missing, the AE model will reconstruct the missing information synchronously, which does not generate meaningful imputation values. When $\beta > 0$, most of the time, the imputation accuracy in the interval of $\beta \in (0, 1)$ is better than that in the interval of $\beta > 1$. This is because in the β -VAE model, the

³ <https://github.com/piupiup/kdd2018/blob/master/>

⁴ <https://github.com/ryancheunggit/kddcup2018-of-fresh-air>

⁵ https://github.com/12190143/KDD_CUP_2018

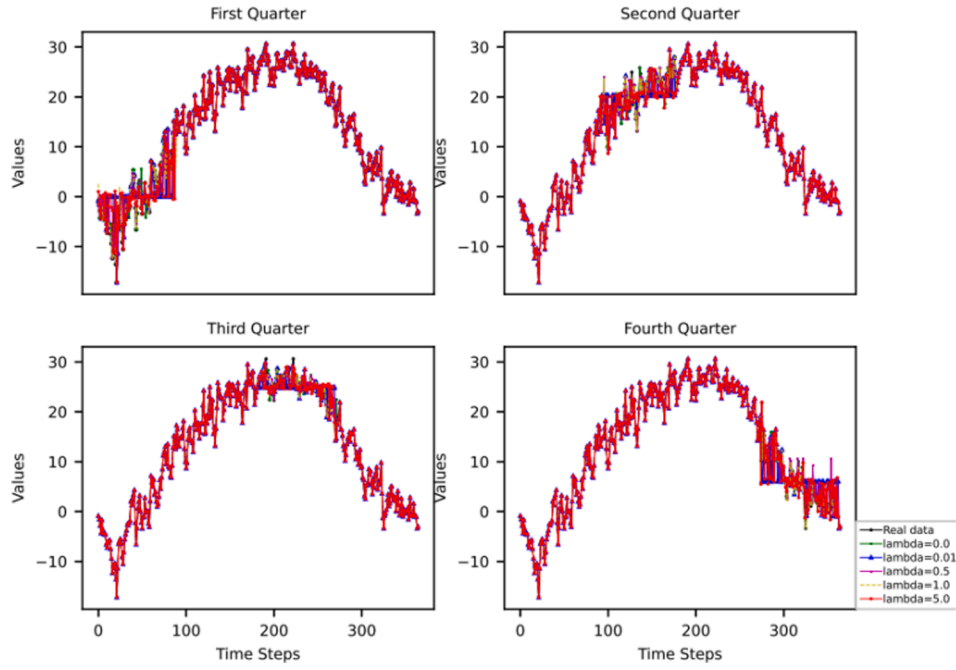


Fig. 7. RMSE performance of specific missing situations on the meteorological data with different λ .

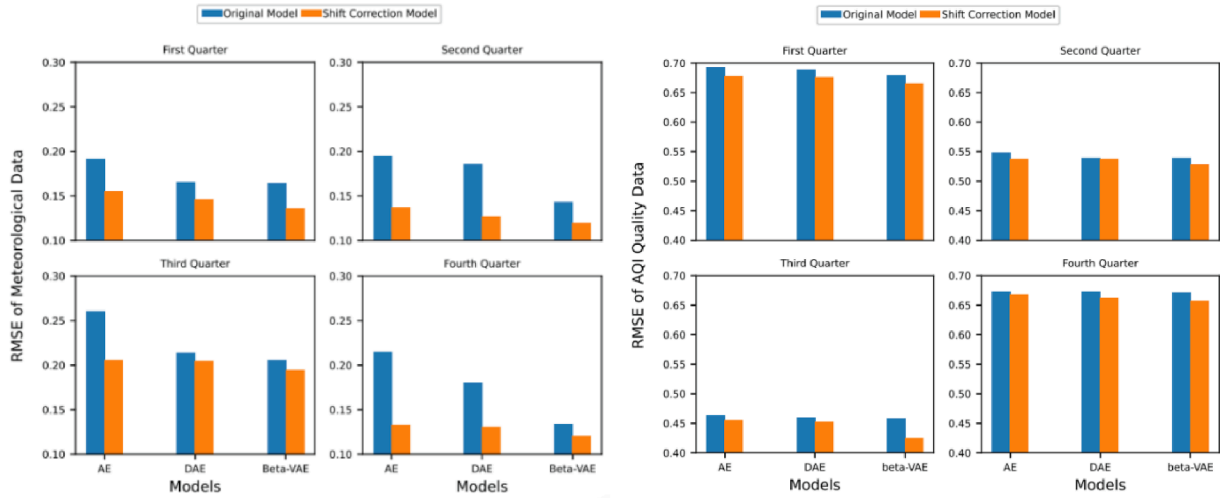


Fig. 8. RMSE performance of specific missing situations on the air quality data with different λ .

ability of generation and disentanglement is balanced and controlled by the hyperparameter β . When $\beta > 1$, the larger β is, the stronger the ability of disentanglement is and the worse the ability of generation is. But as shown in Fig. 4, for the meteorological data during the first quarter and fourth quarter, the imputation effect of the model is better when $\beta > 1$. This may be because missing values in these two quarters belong to the smaller missing situation, which has little impact on the original probability distribution and does not require high ability of generation. And when $0 < \beta < 1$, with the change of β , the ability of disentanglement decreases and the ability of generation increases accordingly. For the VAE model, to solve the problem of data missing imputation, the stronger the ability of generation means the greater probability to generate effective samples, so the range of β should be between (0,1). In addition, for the meteorological and air quality data sets, the centralized missing in the first and the fourth quarter belongs to the small values missing situation and the large values missing situation respectively, and the centralized missing in the second and the third quarter belongs to the large values missing situation and the small values missing situation respectively. And it can be seen from Table 2 that, obviously, the range

of β should be closer to 1 in the case of small values missing situation, while it should be closer to 0 in the case of large values missing situation. Perhaps because the concentrated missing of large values greatly affects the original probability distribution, the imputation model should have stronger ability of generation. The conclusion of selection principle for the hyperparameter β obtained in this subsection will be used in subsequent experiments.

B. The hyperparameter λ . As mentioned in the Section 3, in this paper the hyperparameter λ is used to correct the probability distribution with missing values. It is expected that the effect of probability distribution deviation caused by data concentrated missing can be decreased by shift correction. Then the distribution of train and test sets can be consistent as much as possible, which ensures the basic assumption of the VAE model. This subsection mainly studies the effect of $\lambda \in \{0.0, 0.01, 0.05, 0.1, 0.5, 1.0, 5.0, 10.0\}$ on the imputation performance of specific missing situation and further understands the selection principle of the hyperparameter λ . In the experiment of this subsection, in order to better verify the effect of the hyperparameter λ , the hyperparameter β is set as 1 based on the standard VAE model.

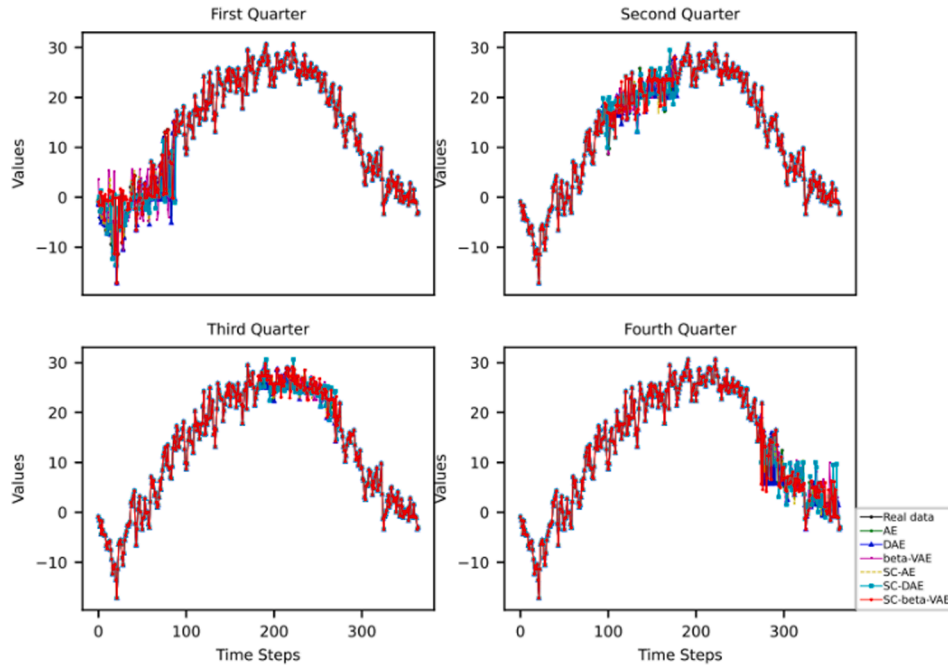


Fig. 9. Imputation results of different λ on the meteorological data.

Table 3 presents results of RMSE with different λ in specific missing situation, which shows the effects of different λ on the imputation accuracy of meteorological and air quality data respectively. And the imputation performance drawn with RMSE as ordinate are shown in Figs. 7 and 8. The time stamps in test set of the meteorological data are regarded as an example, and the imputation results of different β are shown in Fig. 9. When $\lambda = 0$, the imputation model is the standard VAE model. When $\lambda > 0$, compared with the imputation accuracy of $\lambda \in (0,1)$, it is obvious that the model can obtain better imputation effect when $\lambda \geq 1$. This may be because the introduction of the hyperparameter λ is expected to correct the deviated probability distribution by shift correction. When $\lambda \in (0,1)$, the shift correction produced by the model may be not enough to make up for the probability distribution deviation caused by data concentrated missing, and the model still shows the same imputation effect as the VAE model. When $\lambda \geq 1$, the model can effectively modify the deviated probability distribution and make the probability distribution of latent space closer to the original probability distribution, which improves the probability of producing effective samples and achieves the similar effect of the β -VAE model.

But there are some differences for these two hyperparameters. The β -VAE model enhances the ability of generating effective samples by controlling the loss function. The introduction of the hyperparameter λ improves the probability of generating effective samples by modifying the probability distribution of latent space to make it closer to the original probability distribution, so the range of the hyperparameter λ should be greater than 1. To sum up, it can be seen from Figs. 7 and 8

that the range of λ should not be too large in the case of smaller missing situation. While in the case of large values missing situation, the range of λ should be larger to ensure that the shift correction can make up for the impact of data concentrated missing. In addition, the experimental results show that the selection of the hyperparameter λ is not sensitive to specific missing situation. In different types of missing situation, the slight change of λ in a certain range has little effect on imputation results. Therefore, in practical application, due to the problem of unknown probability distribution deviation caused by data concentrated missing, we can first determine the specific missing type through rough selection of λ , and then select a more appropriate λ . The conclusion of selection principle for the hyperparameter λ obtained in this subsection will be used in subsequent experiments.

4.3.2. Effect of the noise and shift correction

A. *Effect of the introduction of noise.* As mentioned in Section 2, the standard AE model lacks the constraint of noise in the training process. Based on the standard AE model, the DAE model introduces noise into the input to destroy the original data, and then uses the corrupted samples with noise to reconstruct clean samples without noise to replace missing values. Based on the standard AE model, the β -VAE model introduces the Gaussian noise constraint into the latent space to make the probability distribution continuous and smooth, so as to produce more accurate samples to replace missing values. The above two models are both based on the standard AE model as the imputation framework, through the introduction of noise in different ways to obtain a

Table 4

Imputation results (in RMSE) of original models of auto-encoder types and shift correction models on different data sets in specific missing situations. (The top is the meteorological data, and the bottom shows the air quality data).

Quarters	Original models			Shift correction models		
	AE	DAE	β -VAE	SC-AE	SC-DAE	SC- β -VAE
First Quarter	0.1919	0.1652	0.1457	0.1550	0.1466	0.1157
	0.6919	0.6875	0.6455	0.6775	0.6757	0.6256
Second Quarter	0.1948	0.1860	0.1734	0.1371	0.1268	0.1203
	0.5483	0.5398	0.5378	0.5378	0.5368	0.5189
Third Quarter	0.2604	0.2135	0.2435	0.2058	0.2047	0.1945
	0.4626	0.4601	0.4272	0.4555	0.4516	0.4055
Fourth Quarter	0.2146	0.1801	0.1629	0.1329	0.1303	0.1203
	0.6731	0.6729	0.6425	0.6678	0.6623	0.6279

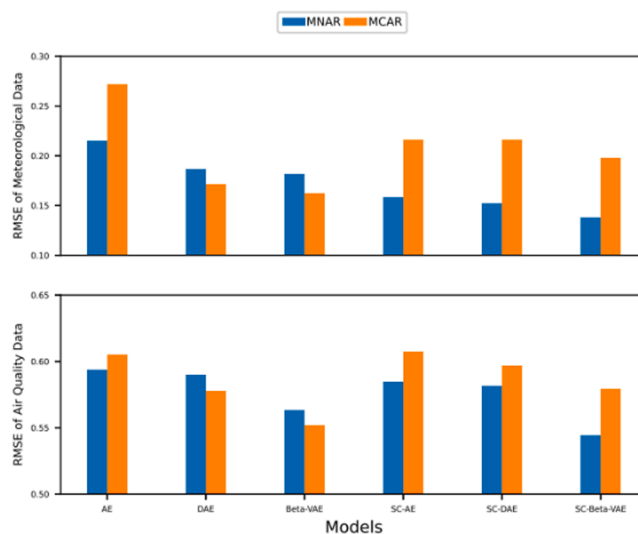


Fig. 10. RMSE performance of original models of autoencoder types and shift correction models on the meteorological data in specific missing situations.

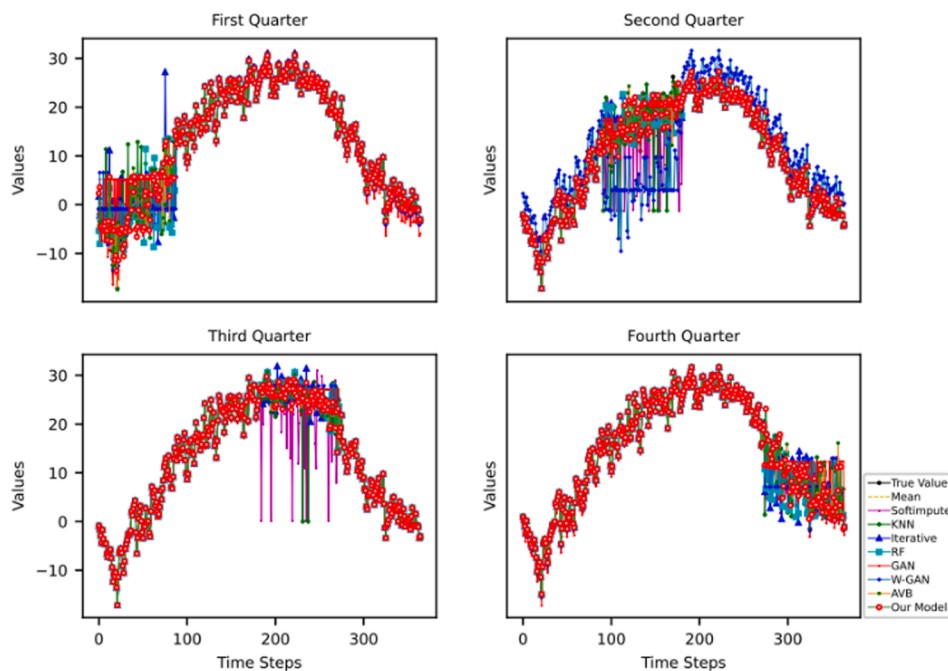


Fig. 11. RMSE performance of original models of autoencoder types and shift correction models on the air quality data in specific missing situations.

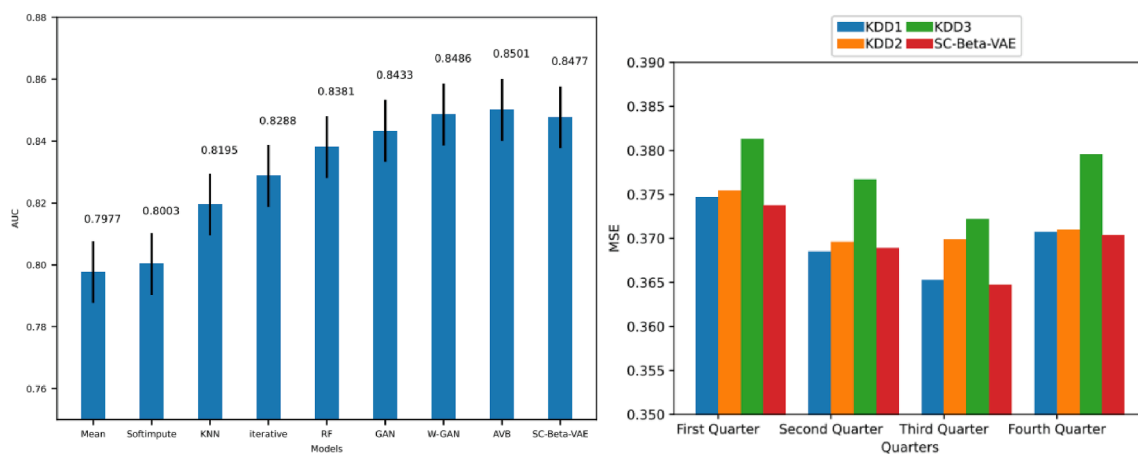


Fig. 12. Imputation results of original models and models after shift correction on the meteorological data.

Table 5

Imputation results (in RMSE) of original models of auto-encoder types and shift correction models on different data sets in different missing situations. (The top is the meteorological data, and the bottom shows the air quality data).

Missing types	Original models			Shift correction models		
	AE	DAE	β -VAE	SC-AE	SC-DAE	SC- β -VAE
MNAR	0.2155	0.1863	0.1814	0.1578	0.1522	0.1378
	0.5940	0.5901	0.5633	0.5847	0.5817	0.5445
MCAR	0.2721	0.1717	0.1627	0.2157	0.2156	0.1982
	0.6051	0.5776	0.5519	0.6077	0.5970	0.5795

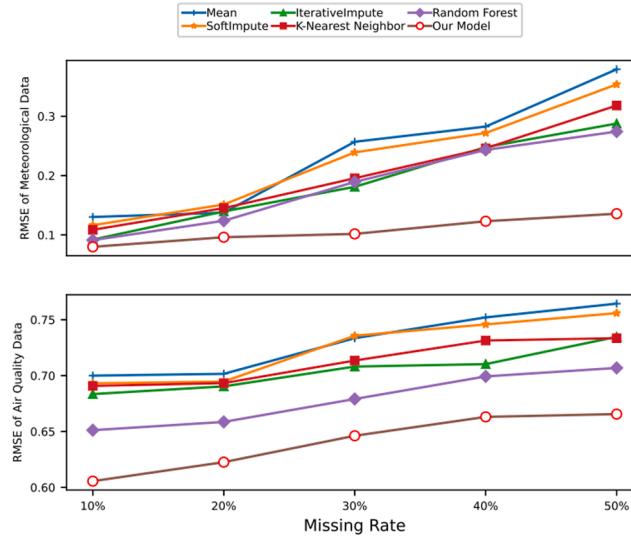


Fig. 13. RMSE performance of original models of autoencoder types and shift correction models on the meteorological and air quality data in different missing situations.

probability distribution that better expresses the real data, which can receive more accurate samples. However, it is different that the results are obtained by introducing noise in different training periods. Therefore, this subsection mainly studies the effect of introduction of noise and different introductions on the imputation performance, and further determines the best training method. In the experiment of this subsection, the hyperparameter β and λ are selected according to the conclusions of Section 4.3.1.

Table 4 presents results of RMSE with different training methods in specific missing situation, which shows the effect of introduction of noise in different training periods on the imputation accuracy of meteorological and air quality data respectively. In order to show the effectiveness of shift correction more intuitively, we compare the effect of original models of auto-encoder types and models after shift correction, and imputation performance is as shown in Figs. 10 and 11. The time stamps in test set of the meteorological data are regarded as an example, and the imputation results of different original models and models after shift correction are shown in Fig. 12. Here, since there is no the latent space in the AE and DAE model, their shift correction is realized by modifying the original probability distribution containing missing data. Obviously, the AE model without noise training and constraints of latent space has the worst imputation results. The DAE model improves the imputation accuracy to a certain extent by destroying the input. And the β -VAE model obtains better imputation results than the DAE model by constraining and controlling the latent space. At the same time, the three models after shift correction all get better imputation results, so the shift correction is effective in specific missing situation. To sum up, in practical application, we should choose the appropriate β and λ for better imputation. By introducing Gaussian noise into the latent space to constrain the original probability distribution, a smoother probability distribution is obtained, thus enhancing the ability of the model to generate effective samples.

B. MNAR VS MCAR. The proposed model mainly aims at the problem of concentrated missing in multivariate time series. It can make up for the impact of concentrated missing on the original distribution by shift correction according to the characteristics of missing data. However, when the data is missing completely at random, the influence on the original distribution is uncontrollable, so the effect of shift correction may be greatly reduced. This subsection mainly shows the performance of models of auto-encoder types before and after shift correction under MNAR and MCAR conditions through experiments, so as to further clarify the application of the proposed method. In the experiment of this subsection, the hyperparameter β and λ are selected according to the conclusions of Section 4.3.1.

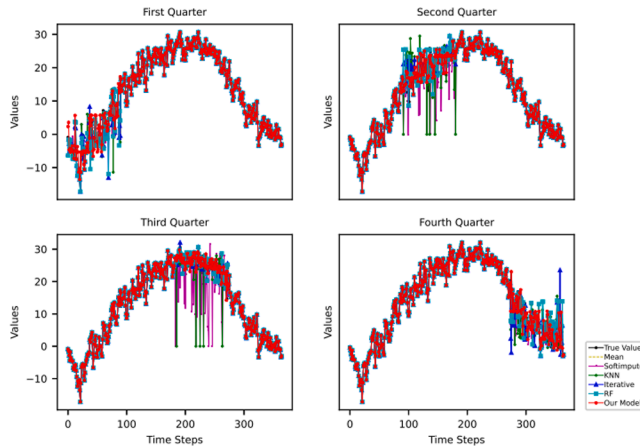
As shown in Table 5, the imputation results of models of auto-encoder types and those with shift correction respectively show the influence of noise and shift correction on the imputation accuracy of meteorological and air quality data under MNAR and MCAR conditions. Among them, the imputation result of MNAR is the average of the model under the condition of concentrated missing in four quarters, and the imputation result of MCAR is the model under the condition of random missing in the whole year. In order to show the imputation effect of the proposed method under different missing conditions more intuitively, the result in Table 5 are plotted as shown in Fig. 13.

From the analysis of the above experimental results, it can be seen that under the condition of MNAR, the shift correction can better improve the impact of concentrated missing on the distribution of original data. However, when the data is MCAR, the accuracy of the imputation model decreases after shift correction, especially for the DAE model and β -VAE model. This may be because the MCAR will not cause a large deviation of the original distribution to a certain extent, and the missing data can be automatically corrected by the variational inference principle. On the contrary, the hyperparameter introduced by shift correction may destroy the learning of original data distribution through

Table 6

Imputation results (in RMSE) of different methods on different data sets in different missing situations.

Quarters	Statistical models			Machine learning models		Generative models			Our model
	Mean	Softimpute	KNN	Iterative	RF	GAN	W-GAN	AVB	
First	0.3791	0.3537	0.3181	0.2877	0.2742	0.2512	0.2348	0.1631	0.1358
	0.7644	0.7559	0.7335	0.7347	0.7069	0.6878	0.6849	0.6724	0.6656
Second	0.3809	0.3968	0.3322	0.3185	0.3065	0.2656	0.2101	0.1686	0.1203
	0.6744	0.6205	0.6070	0.5990	0.5960	0.5733	0.5662	0.5491	0.5290
Third	0.3263	0.3059	0.2986	0.2679	0.2523	0.2370	0.2189	0.1959	0.1945
	0.5109	0.5023	0.5074	0.4991	0.4792	0.4729	0.4687	0.4350	0.4255
Fourth	0.3030	0.2823	0.2919	0.2846	0.2538	0.2485	0.2221	0.1532	0.1204
	0.7199	0.7153	0.7068	0.7073	0.6838	0.6809	0.6727	0.6623	0.6579

**Fig. 14.** Imputation results of different methods on the meteorological data.

variational inference principle. Comparing with the performance of different models under the condition of MNAR and MCAR, it is found that the imputation model before shift correction has better performance under the condition of MCAR, and the imputation model after shift correction is more suitable for the imputation of MNAR.

4.3.3. Evaluation of the proposed model and comparison models on the imputation task

In this subsection, we manually mask 50% data randomly in each specific quarter of the time series to simulate the concentrated missing in the practical application, and these missing values will be as ground truths for verification. Firstly, the imputation performance of the proposed model and comparison models based on statistical theory, machine learning and generative models is tested. This subsection compares the imputation accuracy of different models in specific missing situation and Table 6 shows results of the meteorological and air quality data respectively. Still taking the time stamps in test set of the meteorological data as an example, the imputation results of different models in specific missing situation are shown in Fig. 14.

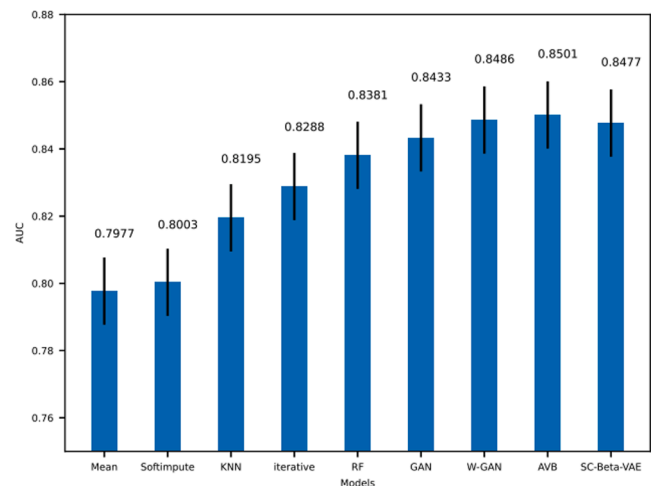
As can be seen from Table 6 and Fig. 14, the imputation method based on statistical theory is worse than that based on machine learning and generative models, and the proposed model can always obtain the optimal imputation results in specific missing situation. This is because the imputation methods based on statistical theory often impute missing values from the perspective of statistical indicators, which does not analyze the characteristics of missing values for specific missing situation. The three imputation methods based on machine learning mainly consider the imputation of missing values from the global perspective, so

the imputation effect is not good in solving specific missing situation. Among the generative models, the AVB model obtains the closest RMSE to the proposed model. This is because the AVB model integrates the adversarial advantages of the GAN model and the variational inference principle of the VAE model, which can obtain more reasonable imputation results. However, in the specific missing situation, the imputation performance of the proposed model is obviously better. Through the proper double control of β and λ , the proposed model can generate more effective samples by the smooth probability distribution of latent space closer to the real data, even for more missing values.

4.3.4. Evaluation of the proposed model and comparison models on the prediction task

In order to further verify the effectiveness of the proposed model in specific missing situation of multivariate time series, two prediction tasks are designed to compare the performance of the proposed model and comparison models.

A. Classification task (mortality prediction task) on Physionet dataset. Because the dataset cannot obtain complete samples, we indirectly compare the imputation performance of the proposed model and comparison models through the accuracy of the prediction task. Firstly, the complete datasets are obtained by imputation of the proposed model and comparison models. And then the classifier model is trained to perform the prediction task based on the complete dataset, so as to indirectly compare the imputation accuracy of these models. Table 7

**Fig. 15.** The AUC score of the mortality prediction trained on different imputation models.**Table 7**

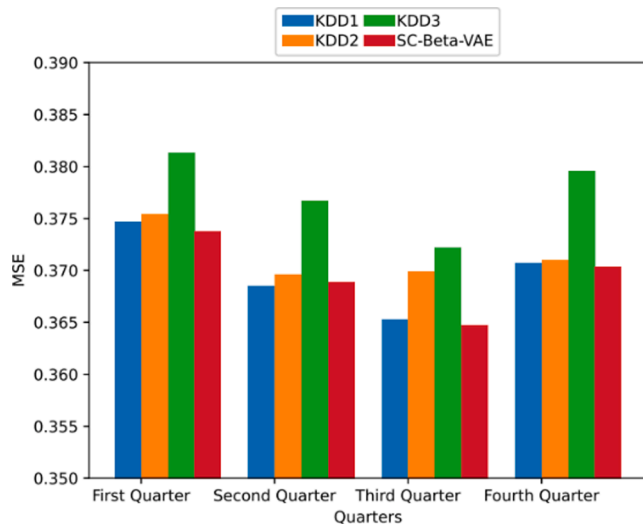
The AUC score of the mortality prediction task on the Physionet dataset.

Statistical models			Machine learning models		Generative models			Our model
Mean	Softimpute	KNN	Iterative	RF	GAN	W-GAN	AVB	
0.7977	0.8003	0.8195	0.8288	0.8381	0.8433	0.8486	0.8501	0.8477

Table 8

Prediction results (in SMAPE) of different models on KDD data set in different missing situations.

Quarters	KDD1	KDD2	KDD3	SC- β -VAE
First Quarter	0.3747	0.3754	0.3813	0.3738
Second Quarter	0.3685	0.3696	0.3767	0.3689
Third Quarter	0.3653	0.3699	0.3722	0.3647
Fourth Quarter	0.3707	0.3710	0.3796	0.3703

**Fig. 16.** The SMAPE score of the air quality prediction trained on different models.

shows the prediction results of the classification task (mortality prediction task) with AUC score as the evaluation index. The prediction performance plotted with AUC score as ordinate is shown in Fig. 15. Combining Table 7 and Fig. 15, since the missing situation of Physionet dataset is MCAR, the classifier based on the original generative models gets better prediction results. However, due to the influence of shift correction hyperparameters, the prediction effect of the proposed model is weaker than that of the generative models on the Physionet dataset with completely random missing.

B. Regression task (air quality prediction task). In this subsection, the KDD dataset is randomly masked by 50% in each quarter to simulate specific missing situation, which are input into the proposed model and comparison imputation models. Then, the imputation results of each model are used as the input of the prediction model. And as shown in Table 8, it is the calculation result of the prediction task with SMAPE as the evaluation index. The SMAPE score for each quarter is the average of all variables. The prediction performance plotted with SMAPE as ordinate is shown in Fig. 16. Combined with Table 8 and Fig. 16, it is obvious that the imputation results obtained from the proposed model can always obtain the minimum SMAPE score in the prediction model. The above results prove that the proposed model has good imputation performance in specific missing situation of multivariate time series.

5. Conclusion and discussion

This paper presents an imputation model for specific missing situations based on the VAE model and shift correction. And the ability of reconstruction of the VAE model is mainly used to build the imputation model. Then the ability of generation is enhanced by the introduction of hyperparameter β , and the problem of the original probability distribution deviation is effectively solved by the introduction of hyperparameter λ . This paper takes the meteorological datasets and some competition data as the example to carry out the experiments of different missing situations, and the following conclusions can be drawn: (1) In different missing situation, by introducing appropriate

hyperparameter β to balance and control the ability of generation and disentanglement, the latent space of the VAE model can fully learn the probability distribution of input, which generates effective samples to replace missing values. And for $0 < \beta < 1$, the experimental results show that the β -VAE model has the stronger ability of generation. (2) By correcting the probability distribution of latent space, the shift correction can effectively weaken the impact of data concentrated missing. The imputation results of specific missing situation are not sensitive to the subtle changes of the hyperparameter λ , and within a certain range it can better make up for the deviation effect caused by concentrated missing. (3) Based on the imputation framework of the standard AE model, the Gaussian noise is introduced into latent space during the training process, which can make the latent probability distribution continuous and smooth and produces more accurate samples. (4) Compared with the typical imputation model based on statistical theory, machine learning and generative models, the proposed model always gets better imputation results in the specific missing situation, which has better robustness.

The above experimental results provide a basis for imputation of specific missing situation in multivariate time series. But some problems should be addressed. On the one hand, for MCAR, the improvement effect of the proposed model is not obvious, and the original imputation model is still recommended. On the other hand, Gaussian noise is introduced into the training of VAE model to constrain the latent space. But the influence of types of noise on imputation results is not considered. Therefore, further research will mainly explore the influence of different types of noise on imputation results in the training of VAE model, and apply the improved model to other real data sets.

CRedit authorship contribution statement

Junying Li: Conceptualization, Methodology, Formal analysis, Software, Writing – review & editing. **Weijie Ren:** Funding acquisition, Resources, Writing – review & editing. **Min Han:** Methodology, Formal analysis, Investigation, Writing – original draft.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Acknowledgments

This work is supported by the National Natural Science Foundation of China (62173063) and the Fundamental Research Funds for the Central Universities (DUT20LAB114).

References

- [1] S. Xiao, J. Yan, M. Farajtabar, et al., Learning Time Series Associated Event Sequences With Recurrent Point Process Networks, *IEEE Trans. Neural Networks Learn. Syst.* (2019) 1–13.
- [2] M. Liu, M. Wang, Q. He, et al., Design and application of time series algorithm model in information assisted sensing system of nursing measurement in neurology, *Measurement* 162 (2020), 107894.
- [3] W. Zhang, X. Li, X. Li, Deep Learning-Based Prognostic Approach for Lithium-ion Batteries with Adaptive Time-Series Prediction and On-Line Validation, *Measurement* 164 (2020), 108052.

- [4] D. Li, M. Han, J. Wang, Chaotic Time Series Prediction Based on a Novel Robust Echo State Network, *IEEE Trans. Neural Networks Learn. Syst.* 5 (23) (2012) 787–799.
- [5] A.J. Zapata-Sierra, A. Cama-Pinto, F.G. Montoya, et al., Wind missing data arrangement using wavelet based techniques for getting maximum likelihood, *Energy Convers. Manage.* 4 (185) (2019) 552–561.
- [6] Y. Li, T. Bao, H. Chen, et al., A large-scale sensor missing data imputation framework for dams using deep learning and transfer learning strategy, *Measurement* 109377 (2021).
- [7] Y.H. Luo, X.R. Cai, Y. Zhang, et al., Multivariate Time Series Imputation with Generative Adversarial Networks, in: *Advances in Neural Information Processing Systems*, 2018, pp. 1596–1607.
- [8] D. Frossard, I.O. Nunes, and R.A. Krohling, An approach to dealing with missing values in heterogeneous data using k-nearest neighbors. *arXiv preprint*, 2016 (arXiv:1608.04037).
- [9] J. Sessa, D. Syed, Techniques to deal with missing data, in: *2016 5th International Conference on Electronic Devices, Systems and Applications (ICEDSA)*, 2016, pp. 1–4.
- [10] S.A. Zahin, C.F. Ahmed, T. Alam, An effective method for classification with missing values, *Applied Intelligence* 48 (2018) 3209C3230.
- [11] Z. Zhang, Missing data imputation: focusing on single imputation, *Ann. Transl. Med.* 1 (4) (2016) 9.
- [12] M. Maniuzzaman, M.J. Rahman, M. Al-Mehedi Hasan, et al., Accurate Diabetes Risk Stratification Using Machine Learning: Role of Missing Value and Outliers, *J. Med. Syst.* 5 (42) (2018) 1–17.
- [13] M. Amiri, R. Jensen, Missing data imputation using fuzzy-rough methods, *Neurocomputing* 205 (2016) 152–164.
- [14] A. Lm, A. Cms, B. Jba, et al., Missing data imputation via the expectation-maximization algorithm can improve principal component analysis aimed at deriving biomarker profiles and dietary patterns, *Nutr. Res.* 75 (2020) 67–76.
- [15] K. Sanjar, O. Bekhzod, J. Kim, et al., Missing Data Imputation for Geolocation-based Price Prediction Using KNNCMCF Method, *ISPRS Int. J. Geo-Inf.* 4 (9) (2020) 227.
- [16] S. Wu, L. Wang, T. Wu, et al., Hankel Matrix Factorization for Tagged Time Series to Recover Missing Values During Blackouts, in: *2019 IEEE 35th International Conference on Data Engineering (ICDE)*, 2019, pp. 1654–1657.
- [17] N. Pauzi, Y. Wah, S. Deni, et al., Comparison of Single and MICE Imputation Methods for Missing Values: A Simulation Study. *Pertanika, Journal of Science and Technology* 4 (29) (2021).
- [18] F. Bießmann, T. Rukat, P. Schmidt, et al., DataWig: Missing Value Imputation for Tables, *J. Mach. Learn. Res.* 20 (2019) 1–6.
- [19] N. Zhou, D. Patel, A. Iyengar, et al., A Verifiable Imputation Analysis for Univariate Time Series and Enabling Package, in: *IEEE International Conference on Big Data*, 2020, pp. 1764–1773.
- [20] I. Izonin, N. Kryvinska, P. Vitynskyi, et al., GRNN Approach Towards Missing Data Recovery Between IoT Systems, *Adv. Intell. Network. Collab. Syst.* (2019) 445–453.
- [21] I. Izonin, R. Tkachenko, V. Verhun, et al., An approach towards missing data management using improved GRNN-SGTM ensemble method, *Engineering Science and Technology, an International Journal* 24 (2021) 749–759.
- [22] R. Tkachenko, I. Izonin, I. Dronyuk, et al., Recover Missing Sensor Data with GRNN-based Cascade Scheme, 2020.
- [23] Y.R. Zhang, Y.L. Zhang, A Comparative Study of Three Multivariate Short-Term Freeway Traffic Flow Forecasting Methods With Missing Data, *Journal of Intelligent Transportation Systems* 3 (20) (2016) 205–218.
- [24] B.K. Beaulieu-Jones, J.H. Moore, Missing data imputation in the electronic health record using deeply learned autoencoders, *Pac Symp Biocomput* 22 (2016) 207–218.
- [25] N. Jaques, S. Taylor, A. Sano, et al., Multimodal autoencoder: A deep learning approach to filling in missing sensor data and enabling better mood prediction, in: *International Conference on Affective Computing & Intelligent Interaction*, 2017, pp. 202–208.
- [26] P.A. Mattei, J. Frellsen, MIWAE: Deep Generative Modelling and Imputation of Incomplete Data, in: *Programs and Abstracts of the International Conference on Machine Learning*, 2018 (4413C4423).
- [27] G. Eraslan, L.M. Simon, M. Mircea, et al., Single-cell RNA-seq de-noising using a deep count autoencoder. *Nature, Communications* 1 (10) (2019).
- [28] A.F. Costa, M.S. Santos, J.P. Soares, et al., Missing Data Imputation via Denoising Autoencoders: The Untold Story. *International Symposium on Intelligent Data Analysis*, 2018.
- [29] U. Garcíarena, R. Santana, An extensive analysis of the interaction between missing data types, imputation methods, and supervised classifiers, *Expert Syst. Appl.* 12 (89) (2017) 52–65.
- [30] S. Ryu, M. Kim, H. Kim, Denoising Autoencoder-Based Missing Value Imputation for Smart Meters, *IEEE Access* 99 (2020), 1–1.
- [31] D. Li, L. Li, X. Li, et al., Smoothed LSTM-AE: A Spatio-Temporal Deep Model for Multiple Time-Series Missing Imputation, *Neurocomputing* 411 (2020).
- [32] Y. Lin, J. Wang, M. Cui, Reconstruction of Power System Measurements Based on Enhanced Denoising Autoencoder, in: *2019 IEEE Power & Energy Society General Meeting (PESGM)*.
- [33] P. Ghosh, M. Sajjadi, A. Vergari, et al., From Variational to Deterministic Autoencoders. *arXiv preprint*, 2019 (arXiv:1903.12436).
- [34] G. Boquet, A. Morell, J. Serrano, et al., A variational autoencoder solution for road traffic forecasting systems: Missing data imputation, dimension reduction, model selection and anomaly detection, *Transp. Res. Part C: Emerg. Technol.* 115 (2020), 102622.
- [35] T. Ilya, B. Olivier, G. Sylvain, et al., Wasserstein Auto-Encoders, 2017 (arXiv: 1711.01558).
- [36] A. Brock, T. Lim, J.M. Ritchie, et al., Neural Photo Editing with Introspective Adversarial Networks. *5th International Conference on Learning Representations*, 2016.
- [37] B. Dai, Y. Wang, J. Aston, et al., Connections with robust PCA and the role of emergent sparsity in variational autoencoder models, *J. Mach. Learn. Res.* 41 (19) (2018) 1–42.
- [38] G. Boquet, J.L. Vicario, A. Morell, et al., Missing Data in Traffic Estimation: A Variational Autoencoder Imputation Method, in: *IEEE International Conference on Acoustics, Speech and Signal Processing*, 2019, pp. 2882–2886.
- [39] V. Fortuin, G. Ratsch, S. Mandt, Multivariate Time Series Imputation with Variational Autoencoders, 2019, pp. 7 (arXiv: 1907.04155).
- [40] Y. Huang, Y.F. Tang, J. Vanzwieten, et al., Reliable machine prognostic health management in the presence of missing data, *Concurr. Comput. Pract. Exp.* 3 (2020), e5762.
- [41] D.P. Kingma, M. Welling, Auto-encoding variational bayes, in: *International Conference on Learning Representations*, 2014.
- [42] A.B. Dieng, Y. Kim, A. Rush, et al., Avoiding latent variable collapse with generative skip models, in: *The 22nd International Conference on Artificial Intelligence and Statistics*, 2019, pp. 2397–2405.
- [43] I. Higgins, L. Matthey, A. Pal, et al., beta-VAE: Learning Basic Visual Concepts with a Constrained Variational Framework, in: *International Conference on Learning Representations*, vol. 5(2), 2017, p. 6.
- [44] C.P. Burgess, I. Higgins, A. Pal, et al., Understanding disentangling in beta-VAE. *arXiv preprint*, 2018 (arXiv:1804.03599).
- [45] D. Rezende, S. Mohamed, Variational inference with normalizing flows, *Computer Science* (2015) 1530–1538.
- [46] S. Ikaro, M. George, J.S. Daniel, et al., Predicting in hospital mortality of ICU patients: The physionet computing in cardiology challenge 2012, in: *Computing in Cardiology (CinC)*, 2012 (245C248).
- [47] D.J. Im, S.J. Ahn, R. Memisevic, et al., Denoising criterion for variational autoencoding framework, *Computer Science*, 2015.
- [48] D. Xu, J.Q. Sheng, J.H. Hu, et al., A Deep Learning Based Unsupervised Method to Impute Missing Values in Patient Records for Improved Management of Cardiovascular Patients, *IEEE J. Biomed. Health. Inf.* 99 (2020) 1.
- [49] I.J. Goodfellow, J. Pouget-Abadie, M. Mirza, et al., Generative Adversarial Nets. *Conference and Workshop on Neural Information Processing Systems*, 2014.
- [50] M. Arjovsky, S. Chintala, L. Bottou, Wasserstein GAN. *arXiv preprint*, 2017 (arXiv: 1701.07875v3).
- [51] L. Mescheder, S. Nowozin, A. Geiger, Adversarial Variational Bayes: Unifying Variational Autoencoders and Generative Adversarial Networks. *arXiv preprint*, 2017 (arXiv: 1701.04722v4).